A Two-Language Replication of
Acemoglu, Johnson, and Robinson (2001),
"The Colonial Origins of Comparative Development"

Replication report

June 9, 2026

Abstract

We replicate the core empirical results of Acemoglu, Johnson, and Robinson (2001, *AER*) — the three scatter relationships (Figures 1–3) and the descriptive, OLS, and instrumental-variables tables (Tables 1, 2, and 4). The replication is carried out *twice*, independently, in Python and in R, reading a single shared copy of the authors' data. Both implementations recover the paper's headline two-stage least-squares (2SLS) estimate of the effect of institutions on income, 0.94 (s.e. 0.16), and agree with each other to the last reported digit. The one substantive cross-language difference we encountered — the standard error of the 2SLS coefficient — turned out to be a degrees-of-freedom convention, which we reconcile so that both languages match the printed paper exactly.

1 The paper in one paragraph

AJR ask whether *institutions* cause long-run prosperity. The empirical problem is reverse causality and omitted variables: rich countries can afford good institutions, so a raw correlation between institutional quality and income is not evidence of a causal effect. AJR's identification strategy uses *settler mortality* as an instrument for institutions. Where European settlers faced high mortality (tropical disease environments) they set up "extractive" states rather than settling; where mortality was low they reproduced European-style institutions protecting property rights. Settler mortality plausibly affects income today *only* through its historical effect on institutions, making it a valid instrument. The causal chain is:

$$\text{settler mortality} \longrightarrow \text{early institutions} \longrightarrow \text{current institutions} \longrightarrow \text{income today.}$$

The three key variables are `logpgp95` (log GDP per capita, PPP, 1995), `avexpr` (average protection against expropriation risk, 1985–95, the institutions proxy), and `logem4` (log settler mortality, the instrument). The main analysis runs on a "base sample" of 64 former colonies.

2 What we replicated

- **Figure 1** — reduced form: income vs. settler mortality.
- **Figure 2** — OLS: income vs. expropriation risk.
- **Figure 3** — first stage: expropriation risk vs. settler mortality.

- **Table 1** — descriptive statistics by sample and mortality quartile.
- **Table 2** — OLS regressions of income on institutions.
- **Table 4** — 2SLS regressions (the headline result), with first stage and OLS panels.

3 Figures

The three figures reproduce the paper's bivariate relationships on the base sample ($n = 64$), plotting country codes as markers with an OLS fit line. The fitted slopes match the corresponding regression coefficients in the tables: the Figure 2 slope (0.522) is the Table 2 column (2) OLS coefficient, and the Figure 3 slope (-0.607) is the Table 4 first-stage coefficient.

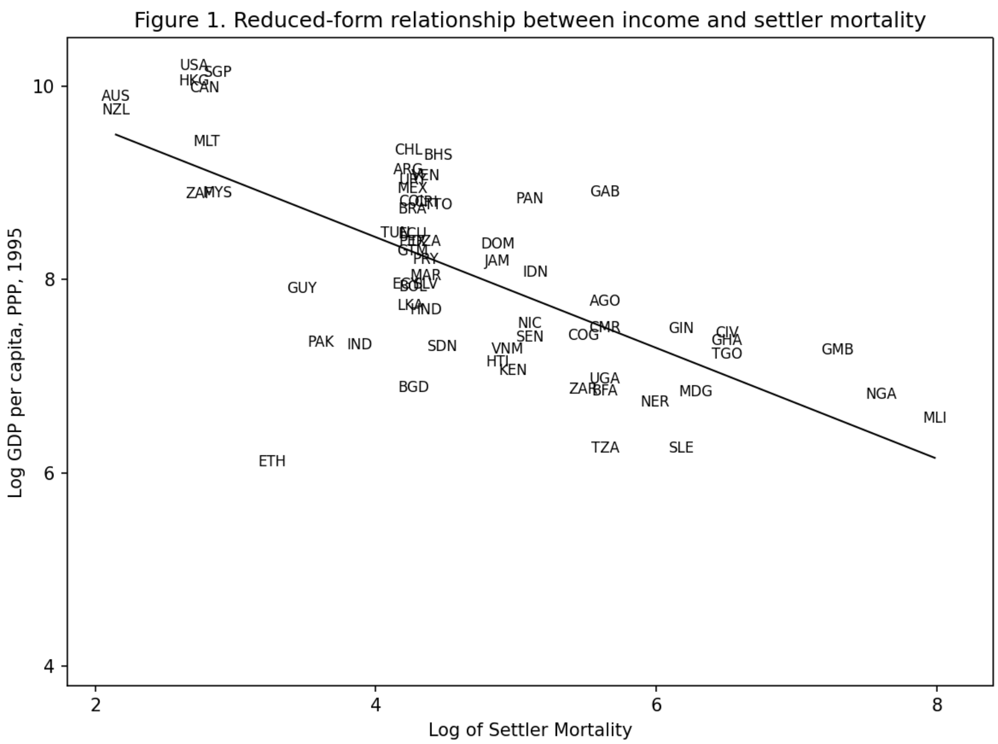

Figure 1: Reduced-form relationship between income and settler mortality (slope -0.573, s.e. 0.076, $R^2 = 0.477$). High-mortality colonies are poorer today.

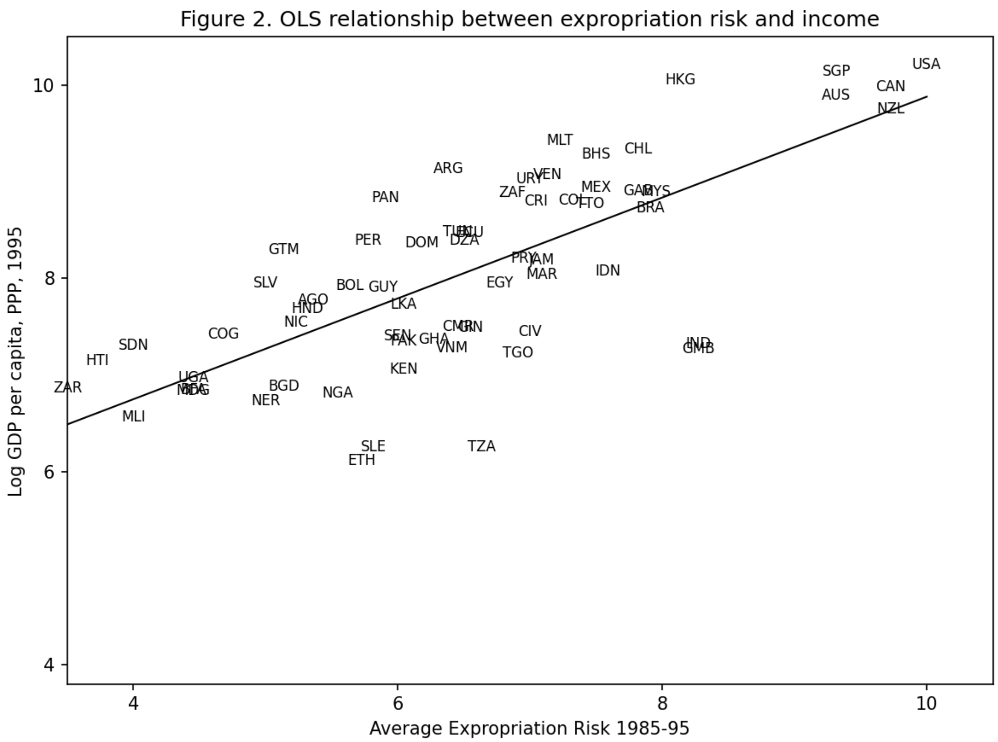

Figure 2: OLS relationship between expropriation risk and income (slope 0.522, s.e. 0.061, $R^2 = 0.540$). Better property-rights protection is associated with higher income.

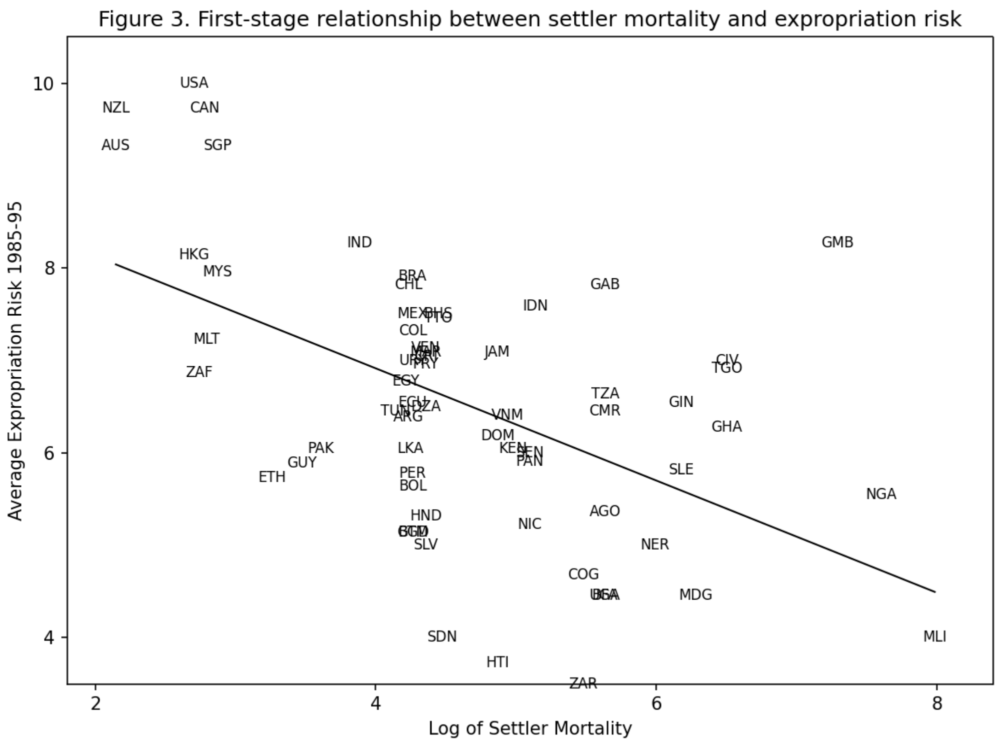

Figure 3: First-stage relationship between settler mortality and expropriation risk (slope −0.607, s.e. 0.127, $R^2 = 0.270$). The instrument is strongly correlated with the endogenous regressor.

4 Tables

All tables report the reproduced values; standard errors are in parentheses. Each matches the printed paper up to rounding. The notes below each table flag the few known data-driven discrepancies.

Table 1: Descriptive statistics (mean; standard deviation in parentheses). Columns (1)–(4) split the base sample by quartiles of settler mortality.

	Whole world	Base sample	(1)	(2)	(3)	(4)
Log GDP per capita (PPP) 1995	8.30 (1.11)	8.06 (1.04)	8.89 (1.29)	8.41 (0.64)	7.78 (0.80)	7.20 (0.63)
Log output per worker 1988	-1.73 (1.08)	-1.93 (0.98)	-1.03 (0.86)	-1.46 (0.42)	-2.20 (0.80)	-3.03 (0.48)
Avg. expropriation risk 1985–95	7.07 (1.80)	6.52 (1.47)	7.90 (1.55)	6.47 (0.95)	5.97 (1.32)	5.89 (1.30)
Constraint on executive 1990	3.64 (2.34)	3.97 (2.26)	5.33 (2.10)	5.12 (1.80)	3.31 (2.18)	2.27 (1.58)
Constraint on executive 1900	1.86 (1.82)	2.25 (2.11)	3.67 (2.99)	3.28 (2.11)	1.12 (0.50)	1.00 (0.00)
Constraint at independence	3.59 (2.41)	3.40 (2.39)	5.17 (2.76)	2.44 (1.92)	3.12 (2.28)	3.43 (2.14)
Democracy 1900	1.15 (2.58)	1.64 (3.00)	3.92 (4.58)	2.76 (2.91)	0.19 (0.40)	0.00 (0.00)
European settlements 1900	0.30 (0.42)	0.16 (0.26)	0.32 (0.44)	0.26 (0.18)	0.08 (0.12)	0.01 (0.02)
Log settler mortality	n.a.	4.66 (1.26)	3.01 (0.59)	4.28 (0.05)	4.92 (0.40)	6.35 (0.75)
Number of observations	163	64	14	18	17	15

Table 2: OLS regressions. Dependent variable is log GDP per capita 1995, except columns (7)–(8) which use log output per worker 1988. "base" = 64-country base sample; "world" = whole-world sample.

	(1) world	(2) base	(3) world	(4) world	(5) base	(6) base	(7) world	(8) base
Avg. expropriation risk	0.53 (0.04)	0.52 (0.06)	0.46 (0.06)	0.39 (0.05)	0.47 (0.06)	0.40 (0.06)	0.45 (0.04)	0.46 (0.06)
Latitude			0.87 (0.49)	0.33 (0.45)	1.58 (0.71)	0.88 (0.63)		
Asia dummy				-0.15 (0.15)		-0.58 (0.23)		
Africa dummy				-0.92 (0.17)		-0.88 (0.17)		
"Other" dummy				0.30 (0.37)		0.11 (0.38)		
R^2	0.61	0.54	0.62	0.72	0.57	0.71	0.55	0.49
Observations	111	64	111	111	64	64	108	61

Note: the whole-world $N = 111$, not the printed 110; AJR's replication code notes one observation was mistakenly dropped from the printed table, so 111 is the corrected figure (matching standard reproductions).

Table 4: IV regressions of log GDP per capita — the headline result. Average expropriation risk is instrumented by log settler mortality. Column (9) uses log output per worker. America is the omitted continent dummy.

	(1) base	(2) base	(3) no Neo-Europes	(4) no Neo-Europes	(5) no Africa	(6) no Africa	(7) + cont. dummies	(8) + cont. dummies	(9) output/ worker
Panel A: Two-Stage Least Squares									
Avg. expropriation risk	0.94 (0.16)	1.00 (0.22)	1.28 (0.36)	1.21 (0.35)	0.58 (0.10)	0.58 (0.12)	0.98 (0.30)	1.11 (0.46)	0.98 (0.17)
Latitude		-0.65 (1.34)		0.94 (1.46)		0.04 (0.84)		-1.18 (1.76)	
Asia dummy							-0.92 (0.40)	-1.05 (0.52)	
Africa dummy							-0.46 (0.36)	-0.44 (0.42)	
"Other" dummy							-0.94 (0.85)	-0.99 (1.00)	
Panel B: First Stage (dependent variable: expropriation risk)									
Log settler mortality	-0.61 (0.13)	-0.51 (0.14)	-0.39 (0.13)	-0.39 (0.14)	-1.21 (0.22)	-1.14 (0.24)	-0.43 (0.17)	-0.34 (0.18)	-0.63 (0.13)
R^2	0.27	0.30	0.13	0.13	0.47	0.47	0.30	0.33	0.29
Panel C: Ordinary Least Squares									
Avg. expropriation risk	0.52 (0.06)	0.47 (0.06)	0.49 (0.08)	0.47 (0.07)	0.48 (0.07)	0.47 (0.07)	0.42 (0.06)	0.40 (0.06)	0.46 (0.06)
Observations	64	64	60	60	37	37	64	64	61

Note: the public `maketable4.dta` does not ship the "Other" continent dummy used in columns (7)–(8); we merge it in from `maketable2.dta` on the country code. Panel A standard errors use the small-sample $(n - k)$ correction, matching the printed paper.

5 Reading the results

The headline comparison is OLS vs. IV. In Table 2 the OLS coefficient on expropriation risk is about 0.52 (base sample); in Table 4 Panel A the 2SLS coefficient is 0.94. The instrumented estimate is *larger*, not smaller. AJR read this as evidence that OLS is, if anything, attenuated by measurement error in the institutions proxy, and that the cross-country correlation is not merely reverse causation. Panel B confirms a strong first stage — log settler mortality enters with a coefficient of -0.61 (s.e. 0.13) in the base sample — so the instrument is not weak. The result is robust across samples: dropping the "Neo-Europes" (columns 3–4) or all of Africa (columns 5–6), or adding continent dummies (columns 7–8), leaves a large, significant effect.

6 The two-language replication

We implemented the entire pipeline twice, independently, against one shared copy of the data (`data/maketable{1,2,4}.dta`).

Tools used in each implementation.

Step	Python	R
Read Stata `.dta`	`pandas.read_stata`	`foreign::read.dta`
Figures	`matplotlib`	base graphics
OLS (Tables 1–2)	`statsmodels`	`stats::lm`
2SLS (Table 4)	`linearmodels.IV2SLS`	`ivreg::ivreg`

The two stacks were chosen to be idiomatic rather than identical. On the R side, figures and the OLS tables need only base R plus the bundled `foreign` package — no installation — while Table 4 adds the lightweight `ivreg` package.

Agreement. Every point estimate agrees between Python and R to the two decimal places reported, and both match the paper. The figures' slopes, R^2, and sample sizes are identical across languages.

The one wrinkle: 2SLS standard errors. Our first R and Python versions disagreed on a single quantity — the standard error of the headline 2SLS coefficient: R reported 0.16, Python 0.15. This was not a bug but a *degrees-of-freedom convention.* R's `ivreg` divides the residual variance by $n - k$ (the textbook small-sample correction AJR used), whereas `linearmodels` with `cov_type="unadjusted"` divides by n. Setting `debiased=True` in the Python fit applies the $n - k$ correction, after which both languages report 0.94 (0.16) — matching the printed paper exactly. This is a useful illustration of why a second, independent implementation is valuable: it surfaced a silent modeling assumption that a single-language replication would never have questioned.

7 Data notes and minor discrepancies

- **Whole-world $N = 111$, not 110.** AJR's own replication code notes one observation was mistakenly dropped from the printed Table 2; 111 is the corrected sample, and matches standard reproductions.
- **The "Other" continent dummy** is absent from the public `maketable4.dta`, so we merge it from `maketable2.dta` on the country code for Table 4 columns (7)–(8).
- **Mortality-quartile boundaries (Table 1).** Five countries are stored at the float value of 78.1; the paper assigns them to quartile (2), so the (2)/(3) cut is placed at 78.15 to reproduce the 14/18/17/15 split.

8 Reproducibility

From the `paper-replication/` directory:

`make python` — run all Python scripts (→ `python/figures`, `python/tables`)
`make R` — run all R scripts (→ `R/figures`, `R/tables`)
`make all` — both

Each language writes its figures as `.png` and its tables as plain-text files; the values in this report are taken directly from those outputs.

Reference: Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The Colonial Origins of Comparative Development: An Empirical Investigation. *American Economic Review*, 91(5), 1369–1401.